Citation Cliques in Questionable Journals

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Abstract

This study examines whether journals of relatively low influence –defined through subject-normalized Eigenfactor percentiles– are associated with denser and more reciprocating patterns of author-to-author citation. Using Crossref records, we assign journals to broad subject areas, compute subject-specific Eigenfactor scores, propagate venue quality to works and authors, match authors in low- versus high-influence venues by subject and h_5, and analyze citation edges for cohesion and anomalies. Across 10,116 matched pairs, authors in low-influence venues exhibit 6.7× higher co-author citation rates and 11× greater clique strength. A subject-aware hybrid detection pipeline achieves 93.5% Case purity (i.e., 93.5% of flagged authors belong to the Bottom tier), revealing a stark "Two Worlds" segregation ($r = 0.71$) where low-influence venues operate as closed citation economies. The largest component ($n = 23$) displays a hub-and-spoke topology, funneling citations to central beneficiaries through coordinated bursts. Overall,

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cohesion –rather than broad asymmetry– accounts for the main case-control differences, suggesting that low-influence venues foster dense, inward-looking citation practices.

Keywords: scientometrics; low-influence journals; citations; cliques; social network analysis; eigenfactor

1 Introduction

Quantitative indicators such as citation counts, the h-index, and journal metrics have become central to the assessment of scholarly impact. These measures offer convenient proxies for influence, but they also create incentives for strategic behavior and manipulation (Biagioli & Lippman, 2020). When deployed uncritically, citation-based indicators risk rewarding volume or visibility over substance, with unintended consequences for research integrity and evaluation systems.

Beyond well-known practices such as excessive self-citation, more coordinated behaviors have emerged. Authors may form tightly connected *citation cliques* –groups that disproportionately reference one another, often within low-influence journals as defined by subject-normalized Eigenfactor percentiles. Such practices distort indicators, undermine trust in the scholarly record, and complicate fair comparison across fields. These dynamics exemplify what has recently been described as *citation orchestration* (Evdaimon, Ioannidis, et al., 2024), where citations serve strategic rather than intellectual functions.

Questionable journals provide fertile ground for these dynamics. Their emphasis on publication volume, rapid turnaround, and relaxed editorial oversight creates environments where abnormal citation patterns can thrive. Systemic distortions such as national *home bias* (Qiu et al., 2025) further interact with venue incentives, amplifying the risk that rankings and impact assessments reflect coordination rather than genuine uptake of ideas.

This study investigates whether questionable journals systematically foster cliques and abnormal citation patterns. Using large-scale bibliographic data, we implement a subject-aware

pipeline that links journal influence to authors, matches Bottom- and Top-tier researchers of comparable productivity (by h_5 index), and analyzes their citation networks for cohesion and anomalies.

We find that extreme anomalies cluster exclusively among Bottom-tier authors in questionable venues: they concentrate in unusually dense, inward-looking structures and form compact outlier groups that set them apart from their Top-tier counterparts. These results provide empirical evidence that venue incentives can shape citation behavior in ways that distort bibliometric indicators. Throughout this paper, authors publishing primarily in low-influence venues are referred to as *cases*, and their matched counterparts in high-influence venues as *controls*.

2 Relevant Work

Prior research has highlighted the fragility of citation-based indicators when used as proxies for scholarly quality. Performance measures not only shape incentives but also invite gaming, from excessive self-citation to coordinated cartels (Biagioli & Lippman, 2020). The wider literature on the "metrics culture" documents how quantitative indicators reshape scholarly communication, creating feedback loops between evaluation systems and researcher behavior. These mechanisms underpin the emergence of citation cartels, inflated impact factors, and other forms of metric manipulation that erode the interpretive value of bibliometric indicators.

Venue-level frameworks were developed to mitigate some of these distortions. The Eigenfactor metric (Bergstrom et al., 2008) introduced a network-based approach that discounts journal self-citations and weights each citation by the prestige of the citing venue. This recursive weighting sought to capture influence rather than popularity and to counteract local citation loops. However, later analyses have shown that such corrections remain sensitive to database coverage, field size, and reference density, indicating that structural biases persist even in sophisticated venue-level measures.

At the author level, organized citation exchanges have been formalized as *citation orchestration* (Evdaimon, Ioannidis, et al., 2024). This framework quantifies concentrated citation flows through indicators such as C/h^2, $A50\%C$, and $A50$, revealing unusually cohesive collaboration networks that may signal collective metric inflation. Earlier studies of citation cartels and excessive self-referencing align with this view, emphasizing how reciprocal ties among small groups of authors can yield disproportionate bibliometric visibility. Such coordinated behaviors blur the line between collaboration and manipulation, challenging the assumption that citation networks represent organic intellectual influence.

Systemic and infrastructural distortions further complicate these dynamics. National *home bias* inflates domestic citation counts, particularly in large scientific systems such as China's (Qiu et al., 2025), while technical manipulation has emerged through "sneaked references" inserted into metadata but absent from article text (Besançon, Cabanac, et al., 2024). Industrial-scale paper mills produce fabricated papers that populate citation networks with artificial linkages (Abalkina et al., 2025). Recent investigations reveal even more elaborate schemes, including the creation of fictitious author identities and peer reviewers to infiltrate legitimate journals. As reported by Naddaf (2025), these fake personas –sometimes maintained across multiple journals– enable mills to generate both papers and favorable reviews, prompting publishers to explore researcher identity verification frameworks and persistent identifier checks such as ORCID trust markers.

Finally, the recognition of complex citation distortions has increased the focus on transparency and reproducibility. The *LLM Guidelines* (LLM Guidelines Consortium, 2024) advocate for documenting decision rules and uncertainty, which is critical as machine learning becomes central to integrity audits. Supporting initiatives promote open data and verifiable auditing to prevent large-scale manipulation. These developments suggest that citation systems are socio-technical infrastructures where credibility relies on both methodological openness and quantitative rigor.

The present research extends this discourse by systematically examining the relationship

between author-level citation behaviors and the influence level of the publication venue, while controlling for individual productivity (h_5 index) and disciplinary norms through a matched case-control design. While previous studies have primarily investigated the orchestration of specific bibliometric indicators (Evdaimon, Ioannidis, et al., 2024) or semantic irregularities within article metadata (J. Liu et al., 2024), this study identifies *structural cohesion*, which manifests as dense local clustering and triadic closure, as the primary diagnostic signature of citation anomalies within low-influence journals. By implementing a subject-aware isolation forest on author-to-author networks across 10,116 matched pairs, the analysis demonstrates that such cliques constitute structural byproducts of venue-level incentives rather than isolated asymmetric flows. Consequently, this approach provides a network-centric diagnostic framework for identifying organized citation practices that evade traditional bibliometric oversight by focusing on network closure patterns rather than simple citation counts.

3 Methodology

Overview. We developed a subject-aware pipeline linking journal quality to author citation behavior. The main steps were: (i) journal subject classification, (ii) computation of subject-specific Eigenfactor scores, (iii) propagation of venue quality to works and authors, (iv) matched pairing of Bottom- and Top-tier authors by productivity, and (v) construction of author-author citation networks with cohesion and anomaly features. Figure 1 illustrates the main analysis subgraph of this pipeline.

To ensure consistency across all stages of the data pipeline, intermediate database layers were validated through *Relational Database Unit* (RDBUnit) testing (Spinellis, 2024), which systematically checks referential integrity, record completeness, and join consistency before statistical analysis.

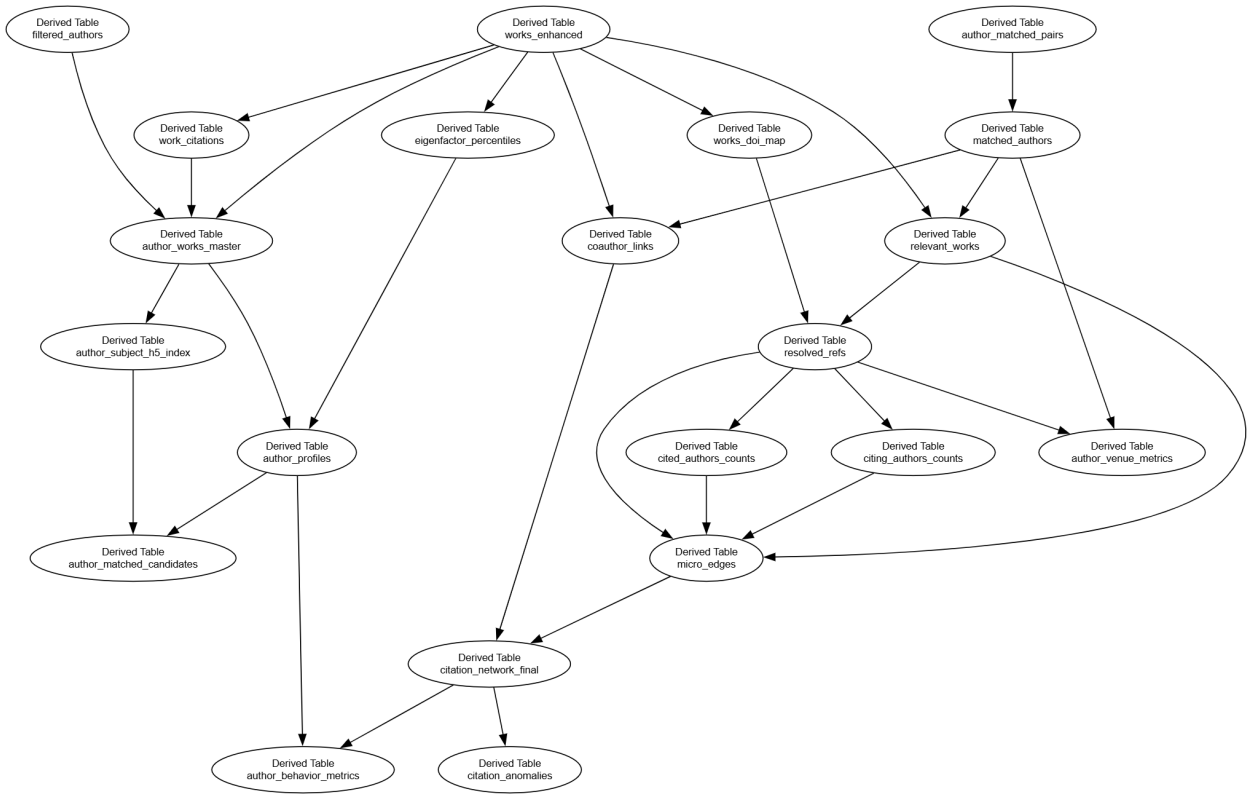

Figure 1: Analysis DAG for the social-network stage. Ellipses represent derived tables and arrows indicate dependencies. Curated inputs feed `citation_network_final` and downstream aggregates.

3.1 Data and Preprocessing

Data collection. Crossref metadata were ingested via *alexandria3k* (Spinellis, 2023), covering works published between 2020 and 2024. We retained DOIs, years, titles, abstracts, references, author identifiers (ORCID), and normalized ISSNs for journals. Duplicates and inconsistencies were cleaned, and derived layers linked works to authors, journals, and citations.

Journal Subject Classification. Each journal was assigned to one of five broad subjects: 1) Health Sciences, 2) Life Sciences, 3) Multidisciplinary, 4) Physical Sciences, 5) Social Sciences & Humanities, using an LLM-based classifier (LLM Guidelines Consortium, 2024). We used `gpt-4.1` with zero temperature and few-shot exemplars, drawing on representative titles and abstracts. Prompts and decision rules prioritized abstracts over titles and journal names. To ensure reproducibility, we used seed and cached predictions to JSON files for

safety.

Eigenfactor Computation. To quantify journal influence, we computed subject-specific Eigenfactor scores (Bergstrom et al., 2008) on journal citation networks (2020-2024), excluding self-citations. The computation used standard power iteration and produced normalized influence scores within each subject.

Defining Venue Quality. Within each subject, journals in the top quartile of Eigenfactor were labeled *high-influence*, and those in the bottom quartile *questionable*. Labels propagate to works (by ISSN) and to authors through their portfolios. Alternative thresholds and persistence checks were evaluated for robustness.

3.2 Author Matching and Network Construction

Author Tiering and Matching. Authors were classified as **Top** if most of their output was in high-influence journals, **Bottom** if concentrated in questionable journals, and **Mixed** otherwise. We then matched Bottom authors to Top authors of similar productivity (subject-specific h_5 index), ensuring comparability in output volume and field.

Sample and Availability. The final analytical dataset includes up to 10,116 matched case-control author pairs per metric (co-author citation rate, clustering coefficient, triangles, maximum and average asymmetry, and average velocity), and 7,711 pairs for maximum citation burst. All pair-level and network-level aggregates were computed per subject area and combined only after validation of within-subject consistency. All processed data tables are available through the public replication repository described in Section 9.

Citation Network and Metrics. For matched authors, we constructed directed author-to-author citation networks (2020-2024). Each reference carried a fractional weight that corrected for team size. From these edges we derived cohesion metrics (self-citation, co-author citation, clustering, triangles, k-core, PageRank) and pairwise anomaly metrics (reciprocity, asymmetry, velocity, citation bursts). These measures capture both dense reciprocal structures and asymmetric one-way flows.

3.3 Statistical Analysis and Outlier Detection

Statistical Analysis and Robustness. Matched pairs were compared using Wilcoxon signed-rank tests (Wilcoxon, 1945), with Benjamini-Hochberg FDR correction (Benjamini & Hochberg, 1995). Effect sizes were quantified via Cliff's δ (Cliff, 1993) and Cohen's d (Cohen, 1988), with confidence intervals estimated through bootstrap resampling (Efron & Tibshirani, 1993). Robustness checks included permutation tests, placebo pairing, random-effects meta-analysis across subjects, and alternative outlier detectors (Isolation Forest (F. T. Liu et al., 2008), Local Outlier Factor, One-Class SVM). Sensitivity analysis utilized z-score thresholds $(1\sigma, 2\sigma, 3\sigma, 4\sigma)$ relative to the control baseline to establish interpretable detection cutoffs. Community detection used the Louvain method (Blondel et al., 2008).

Outlier Detection Procedure. We developed a subject-aware hybrid outlier detection strategy linking multivariate anomalies to specific cohesion signatures. The pipeline combines an Isolation Forest (IF) (F. T. Liu et al., 2008) with a domain-specific Cohesion Composite Score. The IF model used standard hyperparameters (`n_estimators=200`, `contamination=0.01`) to capture general anomalies. The Cohesion Score was computed as the standardized sum of the four most discriminative features: co-author citation rate, clique strength, reciprocity, and outgoing HHI[1]. Feature weights were assigned based on discriminative power (e.g., Co-author Rate weighted 4.0), while authority metrics were inverted. An author was flagged as an outlier only if they satisfied both conditions: an IF anomaly score above the threshold **AND** a Cohesion z-score $> 3\sigma$ (relative to the Control baseline). Sensitivity tests evaluated thresholds from 1σ to 4σ to confirm the robustness of detected syndicates. Outlier labels were merged back into the unified ML table for downstream network analyses (results in Section 4.4).

[1]The Herfindahl-Hirschman Index (HHI) measures concentration. In this context, it quantifies the extent to which an author repeatedly cites the same narrow set of recipients, indicating a lack of source diversity.

4 Findings

This section presents the main results of the matched case-control analysis. We first examine overall and field-specific patterns of cohesion and asymmetry, then derive unsupervised behavioral archetypes and identify statistical outliers using a subject-aware Isolation Forest. Together, these analyses reveal that cohesion-rather than sustained one-sided citation flows-is the primary structural distinction between authors publishing in low- and high-influence venues.

Note on terminology. Throughout this section, terms such as "syndicate," "sycophant," and "beneficiary" are used as *structural labels* that describe observed network roles (e.g., net citation giver vs. net citation receiver). They are adopted for descriptive economy and do not imply intent, coordination, or misconduct on the part of any individual author.

Note on metrics. We summarize the main network features used: the *clustering coefficient* (density of local interconnections); *triangles* (closed three-way citation structures); the *k-core number* (depth of embedding in dense subnetworks); *betweenness centrality* (how often an author lies on shortest paths, a proxy for brokerage/coordination potential); *velocity* (rate of citation accumulation); *burst* (sudden within-window surges of citations); and *asymmetry* (imbalance in giving vs. receiving citations).

4.1 Overall (All Subjects Combined)

Cohesion dominates. At the aggregate level, authors publishing primarily in low-influence venues display markedly stronger *cohesion*: they cite co-authors **6.7× more frequently** ($p = 9.1 \times 10^{-66}$), exhibit **4.7× higher reciprocity rates** ($p = 2.7 \times 10^{-44}$), and show **3.1× greater outgoing concentration** (tunnel vision, $p \approx 0$).

Authority Gap. In contrast to their high internal cohesion, low-influence authors exhibit significantly lower authority metrics. PageRank is **3× lower** for Cases ($p = 3.1 \times 10^{-145}$), and Citation Entropy is reduced by **2.2×**, indicating that while they form tight cliques, they

lack broad authoritative influence.

Net effect. The Bottom-Top gap is explained chiefly by distinct behavioral signatures. These patterns persist under conservative matched-pair comparisons: **12 out of 14** tested behavioral metrics remain significant after Benjamini-Hochberg FDR correction ($\alpha = 0.05$).

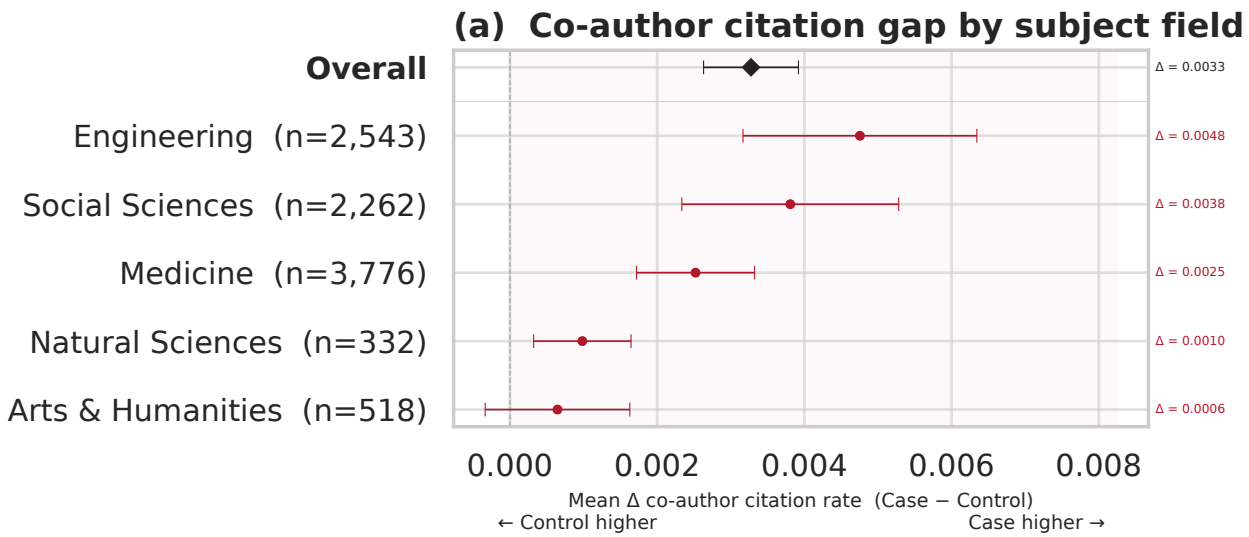

Figure 2: **Co-author citation gap by subject field.** Forest plot of mean paired difference $\Delta = \bar{r}_{\text{Case}} - \bar{r}_{\text{Control}}$ in co-author citation rate, with 95% confidence intervals. Marker • = field estimate; ♦ = overall. Δ labels report rounded effect sizes. The dashed vertical line marks zero (no difference); the pale red shading highlights the Case-higher region.

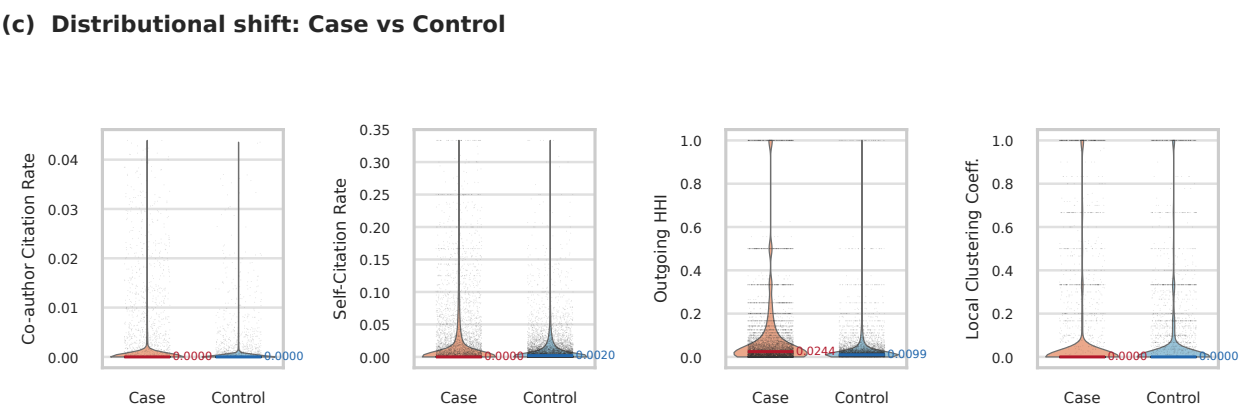

Figure 3: **Distributional Shift: Case vs. Control.** Violin plots for four key metrics: co-author citation rate, self-citation rate, outgoing HHI (tunnel vision), and local clustering coefficient. Cases (red) show systematically heavier right tails than Controls (blue) across all four dimensions, confirming that the cohesion gap is not driven by a few extreme values but by a broad shift in the distribution.

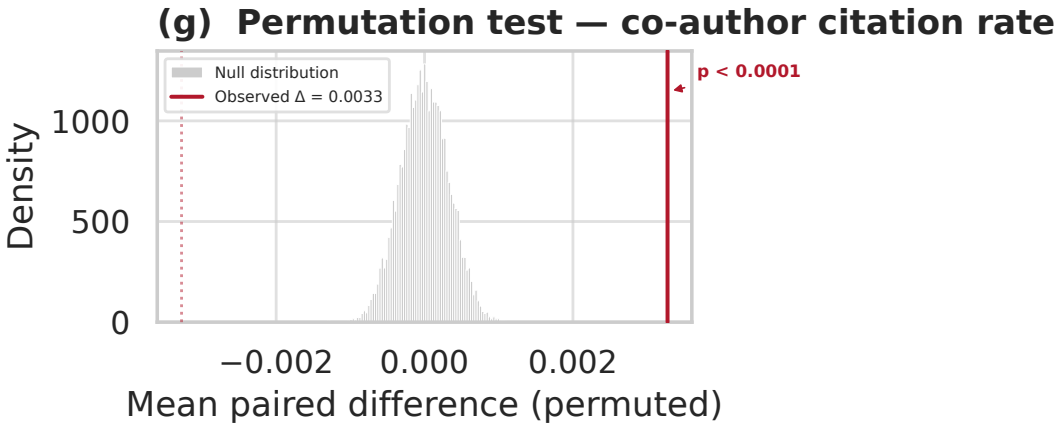

Figure 4: **Permutation Test Results.** The observed difference in co-author citation rates (red line) falls far outside the null distribution generated by 1,000 random permutations ($p = 0.001$), confirming the effect is not due to chance.

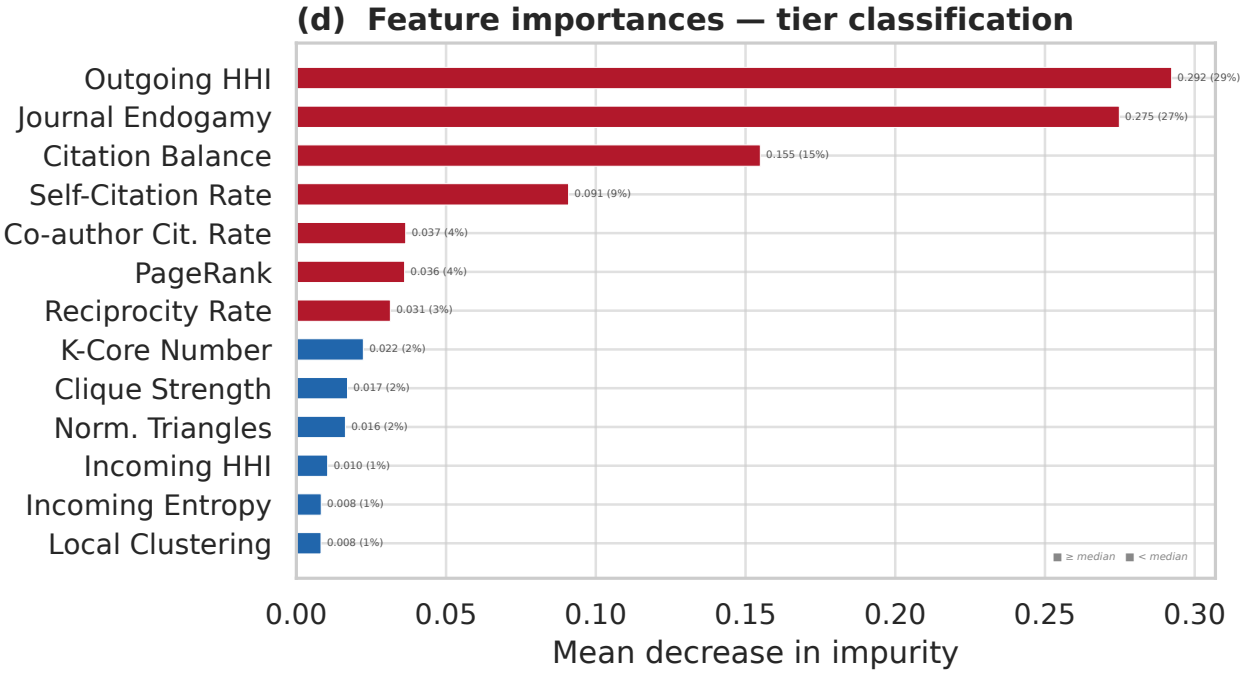

Figure 5: **Random Forest feature importances.** Horizontal bars show mean decrease in impurity for each of the 13 graph-topology features, sorted ascending. Red bars denote features at or above the median importance; blue bars denote below-median. Inline labels give raw importance score and share of total.

4.2 Subject-Stratified Effects (Subjects 1-5)

Robust cohesion, field-specific accents. Enhanced cohesion is a recurrent feature across subjects, but its expression varies by field. In the health and life sciences, the divergence appears mainly as consistently higher co-author citation and stronger local clustering. In multidisciplinary and physical sciences, dense networking more often coincides with short, episodic activity waves. In the social sciences and humanities, cohesion signals are subtler, but sharper temporal concentrations of citations within short windows are more prominent.

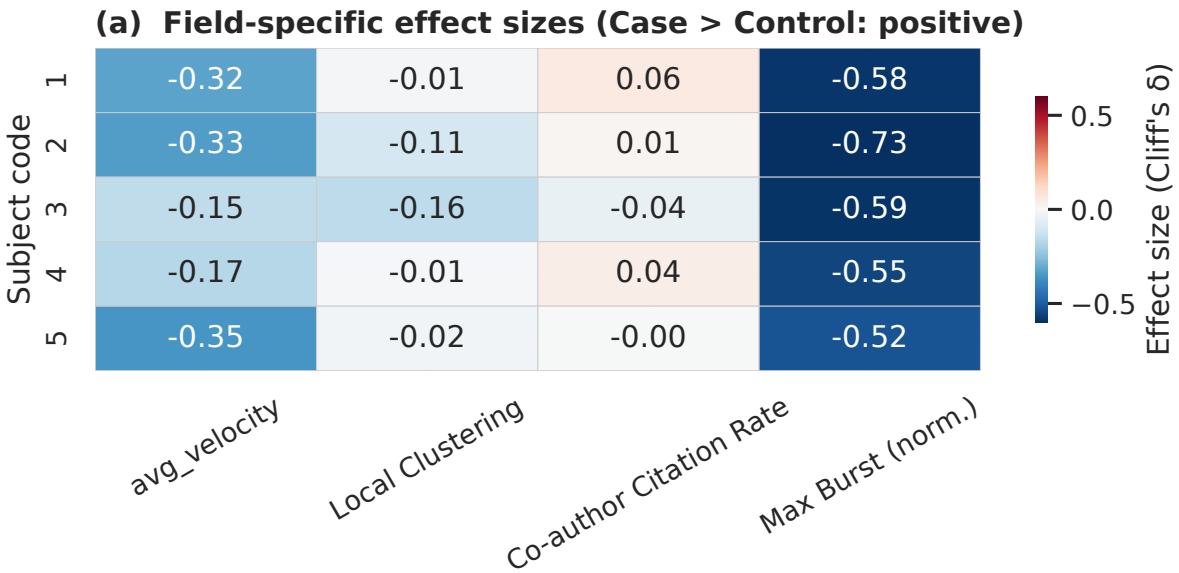

Figure 6: **Subject-Specific Effect Sizes (Cliff's δ).** Each cell reports the effect size for a given metric within a subject area. Positive values (red) indicate that Cases exhibit higher values than Controls. Negative values (blue) indicate that Controls exhibit higher values than Cases. Co-author Citation Rate shows near-zero or slightly positive values across all subjects, while velocity and burstiness are uniformly negative, with Life Sciences (Subject 2) displaying the largest deficit in burstiness ($\delta = -0.73$).

Muted asymmetry. Across fields, one-sided asymmetry measures remain generally subdued. Where differences emerge, they align more with transient speed-ups (velocity/bursts) than with persistent directional imbalance.

Takeaway. The signature of Bottom-tier authors is greater network cohesion. Whether this manifests as intensified co-author referencing, increased local triadic closure, or episodic waves is field-dependent.

4.3 Behavioral Archetypes (Unsupervised)

K-means clustering on standardized ML features reveals a clear three-way typology: **Asymmetric** (unbalanced citing), **Independent** (typical), and **Solo** (self-citing). Bottom-tier authors (Cases) exhibit multiple higher-cohesion strategies, including Asymmetric and Solo behaviors, while Top-tier Controls converge overwhelmingly on the Independent profile. The Solo self-citing pattern appears almost exclusively among Cases.

This divergence is consistent with clique-like behavior: Case authors follow multiple higher-cohesion strategies, whereas Controls converge on a single low-cohesion, outward-looking profile. Figure 7 illustrates this separation: the LDA projection reveals minimal overlap between tiers, with Cases clustering toward the Cohesion-driven extreme and Controls toward the Authority-driven regime.

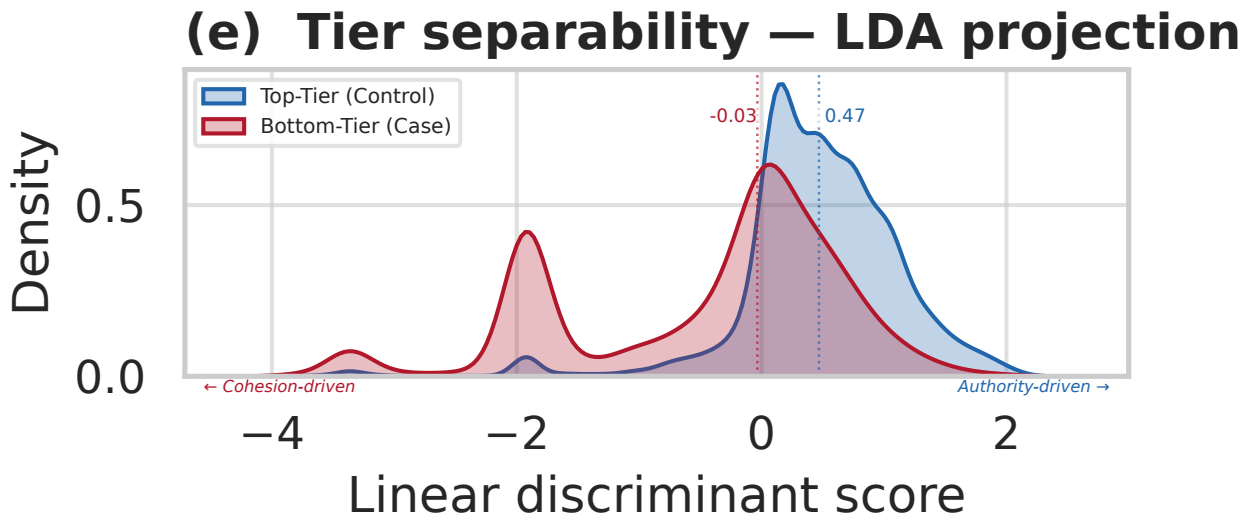

Figure 7: **Tier separability: LDA projection.** Kernel density estimates of the 1-D linear discriminant scores for Bottom-Tier (Case) and Top-Tier (Control) authors. Dotted vertical lines mark within-tier medians. Directional labels indicate the dominant behavioural regime at each tail of the discriminant axis.

Archetype definitions. Three behavioral profiles emerge from the clustering:

- **Asymmetric (Unbalanced)**: high *max_asymmetry* relative to cohesion; concentrated outgoing citation flows.

- **Independent (Typical)**: balanced citing behavior, low cohesion; the dominant profile among Controls.
- **Solo (Self-citing)**: high *self_citation_rate*; nearly absent among Top-tier authors.

The three-way typology provides interpretable separation: *lower-cohesion* profiles dominate among Controls, while *higher-cohesion and asymmetric* profiles are systematically overrepresented among Cases.

4.4 Outlier Detection Results

Detection Accuracy. Using the hybrid detection strategy at the 4σ threshold, we identified *277 high-confidence outliers*. Of these, **93.5%** are Bottom-tier (Case) authors (*Case purity*), confirming that the pipeline predominantly isolates authors whose structural profiles deviate substantially from the Top-tier norm.

Outlier fingerprint. Relative to non-outliers, the detected syndicate members show massive lifts in cohesion signals (Fig. 8): **Clique Strength** $\approx 11\times$, **Co-author Citation Rate** $\approx 6.7\times$, **Reciprocity** $\approx 4.7\times$, and **Tunnel Vision (Outgoing HHI)** $\approx 3.1\times$. These specific deviations constitute the unique *structural fingerprint* of the flagged cohesion outliers.

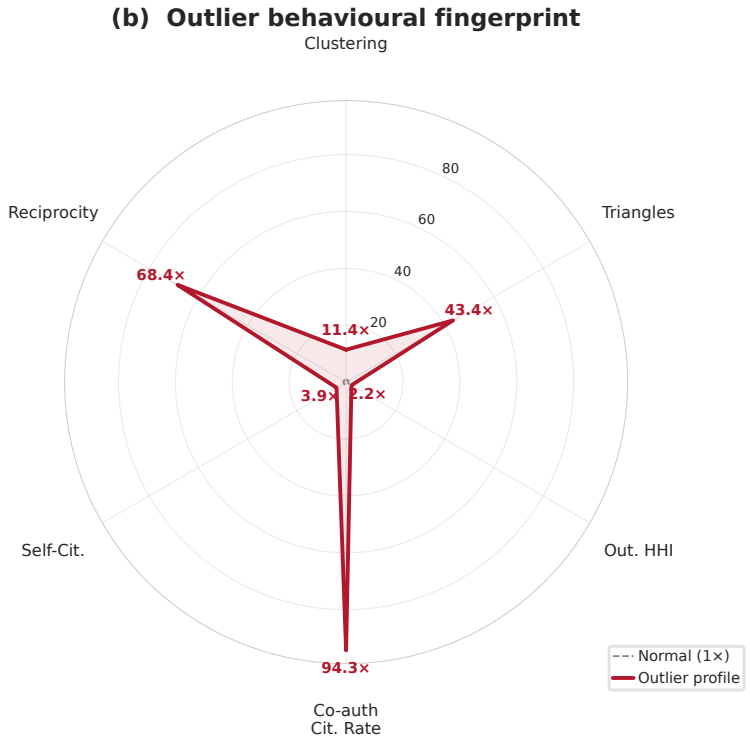

Figure 8: **Outlier behavioural fingerprint.** Radar chart of outlier fold-change ratios relative to normal authors across six citation-behaviour metrics. Values on each spoke indicate the outlier-to-normal mean ratio. The dashed ring marks the baseline (1× normal).

Figure 8 highlights this contrast: outliers broadly exceed non-outliers on cohesion metrics, confirming that the detected anomalies are not random noise but concentrated structural effects.

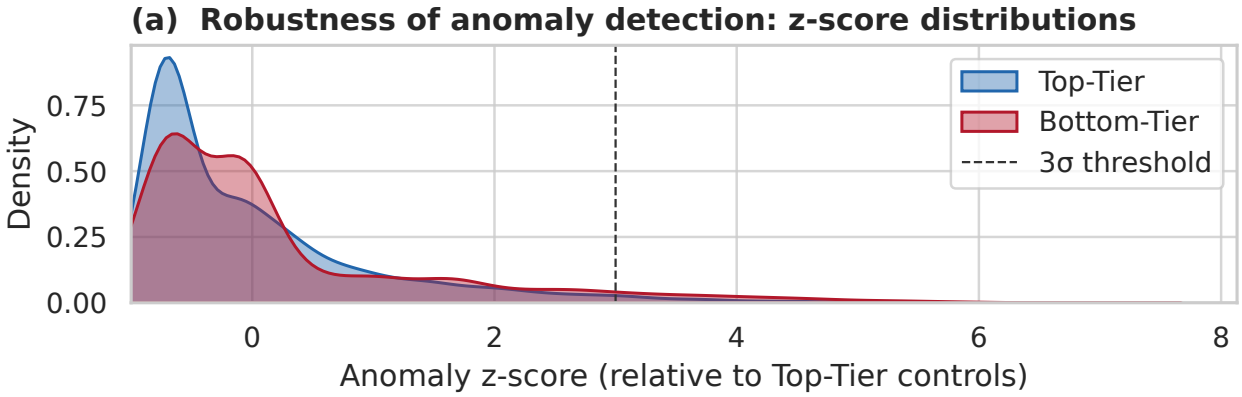

Figure 9: **Robustness of Anomaly Detection.** The distribution of anomaly scores shows a heavy right tail for Cases (Red) extending beyond 3σ. This confirms that the detected outliers represent a distinct structural population, not just normal variability.

Table 1: Sensitivity Analysis: Comparison of Detection Methods at 4σ

Method	**Threshold**	**Outliers**	**Case Purity**	**Connected (%)**
Isolation Forest (Baseline)	4σ	269	84.4%	68.8%
Cohesion Composite Only	4σ	599	89.1%	25.7%
Hybrid (IF $\cap$ Cohesion)	4σ	**277**	**93.5%**	**49.1%**

Note: The Hybrid method combines general multivariate anomalies (IF) with domain-specific cohesion scores, increasing Case Purity by +9.1 percentage points over the baseline.

Table 1 demonstrates the superiority of the hybrid approach. While Isolation Forest alone achieves 84.4% purity, augmenting it with the cohesion filter improves purity to 93.5%. This indicates that approximately 9% of statistical outliers are legitimate prolific collaborators, who are successfully filtered out by the cohesion requirement.

4.5 Network Forensics (Largest Outlier Component)

The largest outlier syndicate ($n = 23$) exhibits a distinct hub-and-spoke topology rather than a dense, uniform mesh. Within this group, the flow imbalance analysis reveals a clear hierarchy: a single **"Primary Beneficiary"** occupies the central position, receiving directed citation flows from peripheral **"Sycophants"** who define themselves by giving significantly more citations than they receive.

Temporal activity supports the orchestration hypothesis, showing a sudden-burst pattern with high internal citation intensity. As shown in the timeline analysis, citation accumulation is not organic but driven by coordinated waves, particularly evident in 2021 and 2024.

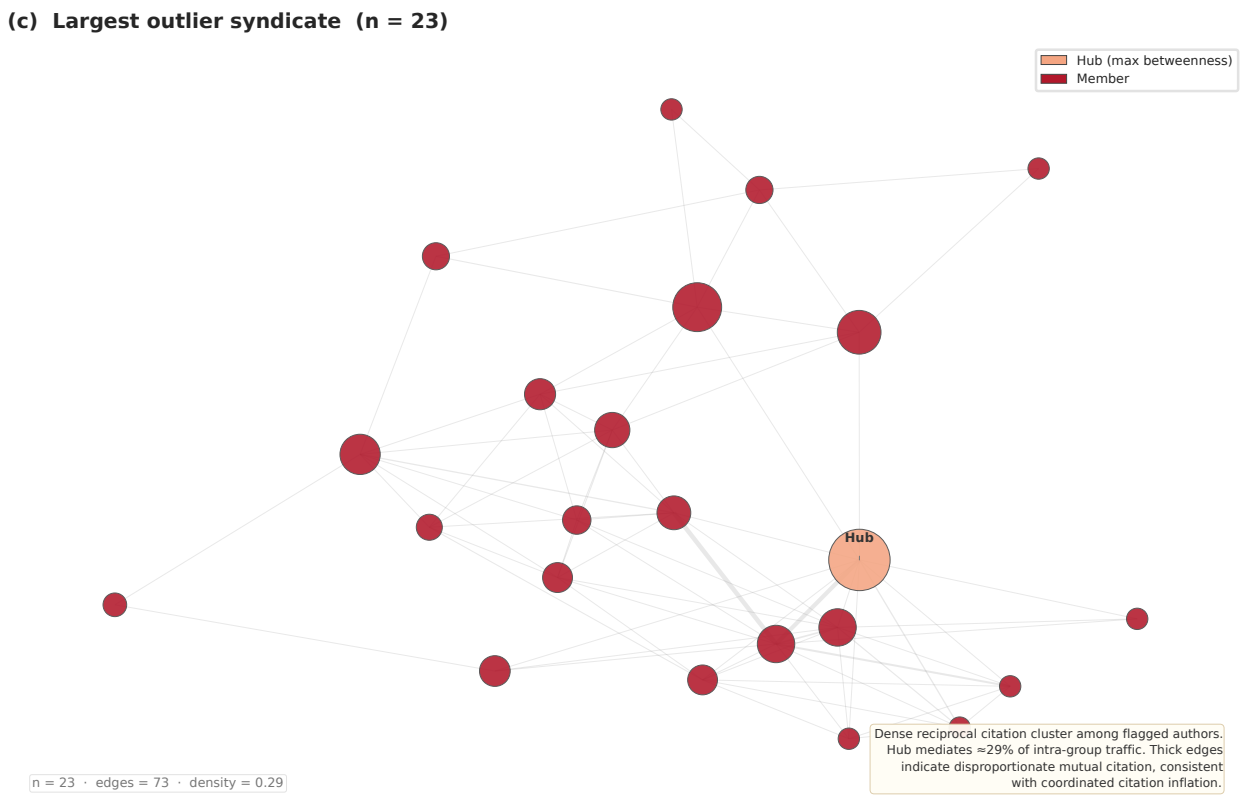

Figure 10: **Largest outlier citation syndicate ($n = 23$).** Force-directed layout of the internal citation network. Node size is proportional to betweenness centrality; salmon = hub (maximum betweenness); red = syndicate member. Edge width scales with aggregate citation weight. The inset box summarises graph statistics.

Figure 10 illustrates the operational mode of the syndicate: temporal bursts coincide with directed citation flows from sycophants to a central beneficiary, revealing a pervasive "Sycophant-Beneficiary" imbalance as a systematic pattern distinguishing Cases from Controls globally.

4.6 Interpretation

Segregation and the "Two Worlds" Phenomenon. The distinction between low- and high-influence venues extends beyond simple density; it reflects extreme *segregation*. The high assortativity coefficient ($r = 0.71$) and modularity ($Q = 0.97$) confirm a **"Two Worlds" hypothesis**, where Bottom-tier authors operate in closed citation economies with minimal

external engagement. Flagged outliers concentrate at the high-asymmetry, high-burst corner of the anomaly landscape, a region essentially unpopulated by Controls. In this sense, citation cliques are not merely organic clusters but isolated silos disconnected from the broader scientific discourse.

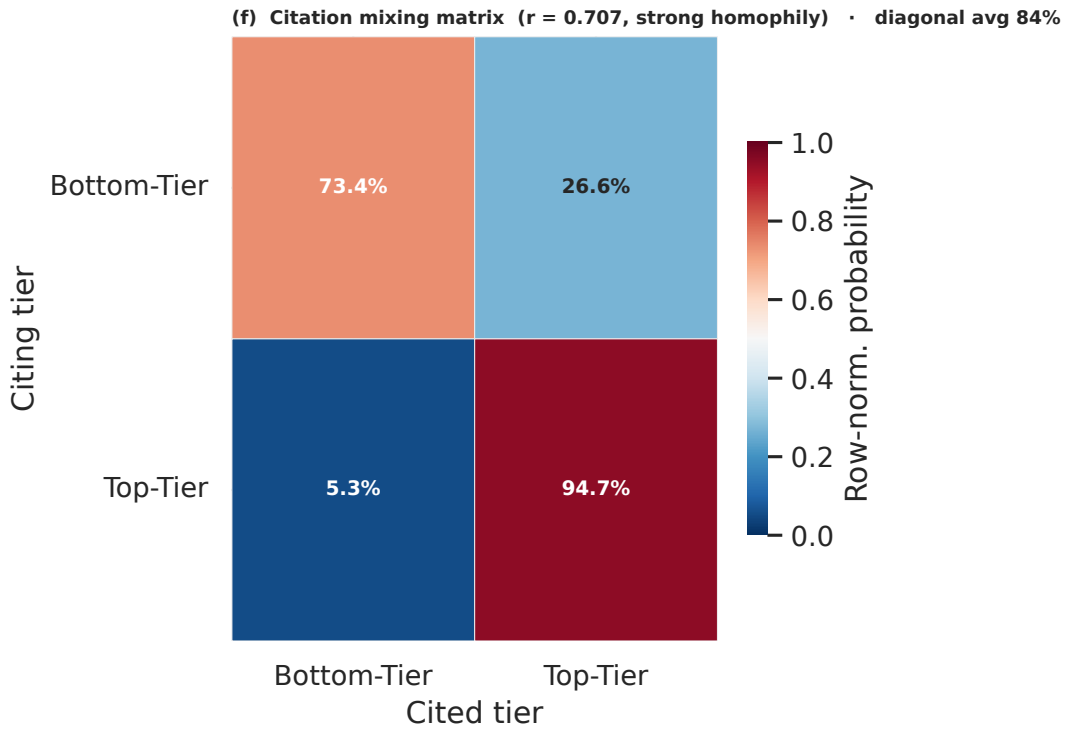

Figure 11: **Citation mixing matrix ($r = 0.71$, strong homophily).** Row-normalised proportion of citations directed within and across tiers. Cell values show the conditional probability that an author in the row tier cites an author in the column tier. The Brewer RdBu colormap is centred at 0.5. Below: diagonal average (84.0%) summarises overall within-tier preference.

Internal Hierarchy: Beneficiaries and Sycophants. While broad asymmetry metrics are generally subdued, the internal structure of detected syndicates reveals a distinct **flow imbalance**. As seen in the largest syndicate ($n = 23$), a hub-and-spoke topology emerges where peripheral members ("Sycophants") act as net givers to central "Beneficiaries". This suggests that the observed cohesion serves a functional purpose: the coordinated funneling of credit to central figures, rather than reciprocal exchange among equals.

Relation to prior literature. These findings extend the concept of *citation orchestration* (Evdaimon, Ioannidis, et al., 2024), showing that orchestration manifests as persistent

reciprocity and dense closure. The limited cross-tier mixing supports concerns about *home bias* (Qiu et al., 2025) and institutional siloing. However, the specificity of our *hybrid detection* (93.5% purity) suggests that these patterns in low-influence venues are structural anomalies distinct from legitimate topic specialization.

Implications for evaluation. The results highlight that aggregate counts hide structural manipulation. Evaluation systems must incorporate **cohesion-aware metrics** –specifically *co-author citation rates* and *flow imbalance*– to distinguish organic collaboration from dense internal referencing. The success of the hybrid pipeline suggests that combining multivariate anomaly detection (Isolation Forest) with targeted cohesion signals can effectively flag insular publication ecosystems.

Limitations. This study relies on Crossref metadata (2020-2024), which vary in completeness. While the hybrid filter significantly reduces false positives, the distinction between a "cartel" and a hyper-specialized research group remains a probabilistic assessment. Moreover, citation links are analyzed structurally, without text-level context to infer the motivation of citing. Finally, the network analysis is conducted at a structural level and does not expose the identities of the individual authors involved; flagged nodes are treated as anonymous structural roles (e.g., "Beneficiary" or "Sycophant") rather than named individuals. Future work could combine the structural fingerprint with author disambiguation and profile-level inspection to move from structural suspicion to targeted auditing.

Future directions. Extending the temporal coverage and triangulating with independent data sources (e.g., OpenAlex) could clarify whether the observed "Two Worlds" segregation is widening over time. Experimental audits in selected venues could also test whether editorial interventions can dismantle the "Echo Chambers" identified in this analysis.

Broader implications. Overall, the dominance of segregation and flow imbalance suggests that low-influence venues foster **closed economies** of citation. Understanding these patterns is essential for designing fair bibliometric indicators that reward genuine impact rather than inward-looking visibility loops.

5 Summary and Conclusions

We developed a subject-aware hybrid pipeline that integrates LLM-based journal classification, per-subject Eigenfactor scores, and a composite outlier detection strategy. This design allowed us to isolate specific behavioral signatures of manipulation once productivity (by subject-specific h_5) is held constant.

The evidence points to a highly structured phenomenon. Authors publishing predominantly in low-influence journals cite co-authors **6.7× more frequently** and exhibit **11× greater clique strength** compared to their top-tier counterparts. Rather than random noise, outlier syndicates display a **hub-and-spoke topology**, where peripheral "Sycophants" funnel citations to central "Beneficiaries" through coordinated bursts. The flow imbalance analysis confirms that Case authors systematically give more citations than they receive. Critically, as shown in Figure 12, this gap is **not converging**: Top-Tier authors accumulate citations at an accelerating rate relative to Cases, confirming that the "Two Worlds" segregation is structurally entrenched and widening over time.

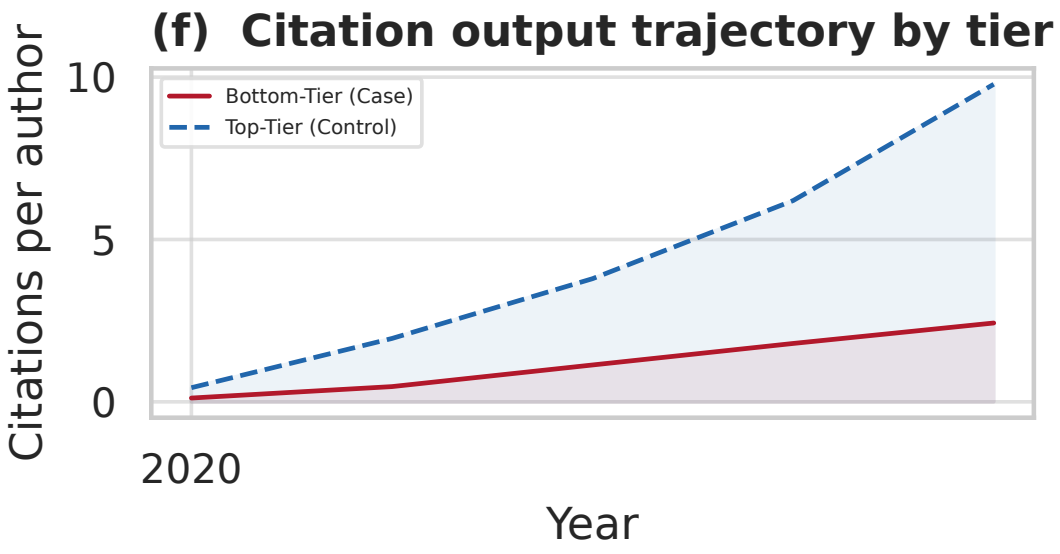

Figure 12: **Citation Output Trajectory by Tier (2020-present).** Cumulative citations per author for Bottom-Tier Cases (solid red) and Top-Tier Controls (dashed blue), with shaded confidence bands. The diverging trajectories confirm that the influence gap between tiers is *widening*, not converging, over time.

Conclusion. Controlling for subject and productivity, the hybrid detection strategy identifies structurally anomalous citation groups with 93.5% Case purity (proportion of flagged authors belonging to the Bottom tier). These groups are characterized not by simple asymmetry, but by a specific cohesion fingerprint –extreme co-author citation rates, dense local clustering, and tunnel vision. Consequently, low-influence venues do not merely host lower-quality work; they foster **segregated, inward-looking citation economies** that actively distort bibliometric indicators through structural manipulation, a divide that is deepening with each passing year.

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7 Author Contributions

Conceptualization: P.-A.S., D.S.; Methodology: P.-A.S.; Software: P.-A.S., G.A.; Data curation: P.-A.S., G.A.; Validation: P.-A.S., G.A., D.S.; Formal analysis: P.-A.S., G.A.; Visualization: P.-A.S.; Writing - original draft: G.A.; Writing - review & editing: P.-A.S., G.A., D.S.; Supervision: D.S.; Project administration: D.S.

8 Competing Interests

The authors declare no conflict of interest.

9 Data and Code Availability

All essential data and code for this study are openly available at Zenodo under a CC BY license (Spanakis et al., 2026).

References

Abalkina, A., Aquarius, R., Bik, E. M., Bimler, D., Bishop, D., Byrne, J., Cabanac, G., Day, A., Labbé, C., & Wise, N. (2025). Stamp out paper mills. *Nature*, *637*, 1047–1049. https://doi.org/10.1038/d41586-025-00212-1

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, *57*(1), 289–300.

Bergstrom, C. T., West, J. D., & Wiseman, M. A. (2008). The eigenfactor metrics: A network approach to assessing scholarly journals. *Journal of Neuroscience*, *28*(45), 11433–11434. https://doi.org/10.1523/JNEUROSCI.0003-08.2008

Besançon, L., Cabanac, G., et al. (2024). When scientific citations go rogue: Uncovering sneaked references. *Journal of the Association for Information Science and Technology*. https://doi.org/10.1002/asi.24896

Biagioli, M., & Lippman, A. (Eds.). (2020). *Gaming the metrics: Misconduct and manipulation in academic research*. MIT Press. https://doi.org/10.7551/mitpress/11087.001.0001

Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, (10), P10008.

Cliff, N. (1993). *Dominance statistics: Ordinal analyses to answer ordinal questions*. Psychological Assessment Resources.

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd). Routledge.

Efron, B., & Tibshirani, R. J. (1993). *An introduction to the bootstrap*. Chapman & Hall.

Evdaimon, I., Ioannidis, J. P. A., et al. (2024). Metrics to detect small-scale and large-scale citation orchestration. *arXiv Cornell University*. https://doi.org/10.48550/arXiv.2406.19219

Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2008). Isolation forest. *Proceedings of the 8th IEEE International Conference on Data Mining*, 413–422.

Liu, J., Bai, X., Wang, M., Tuarob, S., & Xia, F. (2024). Anomalous citations detection in academic networks. *Artificial Intelligence Review*, *57*(4), 103. https://doi.org/10.1007/s10462-023-10655-5

LLM Guidelines Consortium. (2024). *Llm guidelines: Transparency in llm-assisted research*. Retrieved September 17, 2025, from https://llm-guidelines.org/

Naddaf, M. (2025). How to spot fake scientists and stop them from publishing papers [News Feature, 22 October 2025]. *Nature*, *646*, 792–794. https://doi.org/10.1038/d41586-025-03341-9

Qiu, S., Steinwender, C., & Azoulay, P. (2025). *Paper tiger? chinese science and home bias in citations* (NBER Working Paper No. 32468) (Revised from May 2024 version). National Bureau of Economic Research. https://doi.org/10.3386/w32468

Spanakis, P.-A., Alexandrou, G., & Spinellis, D. (2026). *Replication data for: Citation cliques in questionable journals* (Version v1). Zenodo. https://doi.org/10.5281/zenodo.xxxxxxx

Spinellis, D. (2023). Open reproducible scientometric research with alexandria3k. *PLoS ONE*, *18*(11), e0294946. https://doi.org/10.1371/journal.pone.0294946

Spinellis, D. (2024). Unit tests for sql. *IEEE Software*, *41*(1), 31–34. https://doi.org/10.1109/MS.2023.3328788

Wilcoxon, F. (1945). Individual comparisons by ranking methods. *Biometrics Bulletin*, *1*(6), 80–83.